

Article

Effect of Dietary Inclusion of Olive Leaves and Olive Pulp on the Oxidative Status and Meat Quality of Broiler Chickens

Vida Rezar , Manca Pečjak Pal, Janez Salobir  and Alenka Levart 

Department of Animal Science, Biotechnical Faculty, University of Ljubljana, 1000 Ljubljana, Slovenia; manca.pecjak@bf.uni-lj.si (M.P.P.); janez.salobir@bf.uni-lj.si (J.S.); alenka.levart@bf.uni-lj.si (A.L.)

* Correspondence: vida.rezar@bf.uni-lj.si

Abstract: The production of olive oil results in various by-products such as olive leaves and olive pulp, which can be utilized in animal nutrition. The aim of the present study was to investigate the effects of dietary olive leaves and olive pulp on the oxidative status and fatty acid (FA) composition of broiler breast meat. A total of 120 one-day-old male Ross 308 broilers were randomly divided into 5 experimental groups: 1 control group (Cont) without supplementation and 4 experimental groups supplemented with either 5% or 10% olive leaves (OLeav5; OLeav10) or olive pulp (OPulp5; OPulp10). Blood and breast muscle samples were taken at the end of the experiment. The results showed that the addition of olive leaves or olive pulp did not significantly alter the concentration of malondialdehyde (MDA) and the antioxidant capacity of lipid-soluble compounds (ACL) in the blood or the enzyme activities of the liver. However, the antioxidant capacity of water-soluble compounds (ACW) in serum was reduced in broilers receiving 5% olive pulp or 10% olive leaves ($p = 0.002$). In addition, meat quality parameters were not affected by olive leaves or pulp intake, although 10% olive leaves reduced lightness (L^*) ($p = 0.023$) and α -tocopherol concentration in breast muscle ($p = 0.001$) compared to control. Olive leaves and pulp intake also affected the FA profile of the breast muscle, with 5% olive pulp increasing monounsaturated fatty acid (MUFA) content ($p = 0.002$), while 10% olive leaves increased polyunsaturated fatty acid (PUFA) content ($p = 0.015$). In conclusion, supplementation with up to 5% olive leaves or pulp had no adverse effects on the oxidative status and meat quality of broilers.

Keywords: broiler; olive leaves; olive pulp; oxidative status; breast meat



Academic Editors: Petru Alexandru Vlaicu, Mihaela Dumitru and Alexandra Gabriela Oancea

Received: 10 March 2025

Revised: 18 March 2025

Accepted: 18 March 2025

Published: 20 March 2025

Citation: Rezar, V.; Pečjak Pal, M.; Salobir, J.; Levart, A. Effect of Dietary Inclusion of Olive Leaves and Olive Pulp on the Oxidative Status and Meat Quality of Broiler Chickens. *Agriculture* **2025**, *15*, 662. <https://doi.org/10.3390/agriculture15060662>

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

In poultry production, birds are exposed to a variety of stressors, including high ambient temperatures, increased stocking density, disease risks, transportation, different nutritional stressors, and others that negatively impact bird health and productivity [1]. Stressful conditions increase the production of free radicals and lead to oxidative stress, which triggers lipid peroxidation in body tissues [2]. To enhance antioxidative activity and reduce oxidative damage in vivo and consequently in animal products, supplementation with various antioxidants such as vitamins E and C is used effectively [3–5]. In line with the increasing “zero waste” policy and consumer demands for the use of natural ingredients in animal feed, there is growing interest in implicating natural antioxidants such as polyphenol-rich plant extracts and herbs [6]. Furthermore, feed costs account for about 70 percent of total costs in poultry production. The ongoing feed-food competition and the increasingly higher costs of key feed ingredients such as grains and oilseeds highlight the

need to find potential natural feed sources. The use of industrial or agricultural by-products could help to reduce overall production costs while maintaining the nutritional value of the feed [7].

Olive oil is widely recognized as a key component of the Mediterranean diet and is valued for its health-promoting properties due to its high content of monounsaturated fatty acids (MUFAs) and polyphenolic compounds [8]. However, olive oil production generates significant by-products, including olive leaves, pomace, wastewater, and stones, which can contribute to environmental pollution if not managed properly due to their high organic content and phytotoxic properties [9]. It is estimated that olive leaves account for about 10% of the total weight of harvested olives, with each olive tree generating around 25 kg of leaves annually from pruning. Furthermore, producing 200 kg of extra virgin olive oil generates up to 800 kg of pomace and 200 kg of wastewater [10]. To mitigate waste accumulation and reduce disposal costs, the integration of these by-products into animal nutrition represents a sustainable alternative that aligns with the principles of the circular economy [11].

In olive oil extraction, olive pulp (OPulp) is the residue obtained when the olive stones are separated from the crude olive cake, while the olive leaves (OLeav) are obtained when the olive trees are pruned and the olive fruits are cleaned before extraction [11]. Due to their bioactive compounds, olive by-products have a high potential for use in the food and pharmaceutical industries and as feed additives in animal nutrition [12]. Phenolic compounds in olive oil and its by-products, such as oleuropein, hydroxytyrosol, tyrosol, and others, exhibit strong antioxidant activity, act as free radical scavengers, and are thought to protect the cardiovascular system and reduce the risk of certain diseases [13]. Consequently, both OLeav and OPulp possess several antioxidant, anti-inflammatory, anti-hypertensive, antimicrobial, antibiotic, and antithrombotic properties [9]. The antioxidant properties of olive oil polyphenols have also been recognized by the European Food Safety Authority [14] through a health claim associated with the protection of body cells and low-density lipoproteins (LDL) from oxidative damage.

When using olive by-products in poultry diets, their nutritional limitations must also be considered. Although they are a good source of protein, fat, and some minerals, e.g., calcium, copper, and cobalt, they have a low nutritional value due to their low energy content, some minerals (phosphorus, magnesium, sodium) and digestible protein content, as well as the high lignin content [15]. Furthermore, higher levels of olive by-products can negatively affect nutrient absorption and digestibility and impairs body weight gain (BWG) [16]. This effect is likely due to the tannins and non-starch polysaccharides (NSP) present in the olive plant's cell walls [17]. However, various studies reported that the including dried olive by-products in broiler diets at levels of 5 to 10% does not negatively impact growth performance, carcass traits, or histological and blood parameters [7,18–20]. Research on the antioxidant benefits of olive by-products in poultry is relatively limited. Nonetheless, supplementing broiler feed with olive leaf extract has demonstrated notable benefits as it can enhance the birds' antioxidant capacity and gut health, and it reduces lipid oxidation in blood plasma and breast meat, evidenced by lower malondialdehyde (MDA) [21,22]. Similarly, Gerasopoulos et al. [23] reported that supplementation with olive mill wastewater decreased protein and lipid oxidation and increased antioxidant capacity in plasma and tissues of broilers. Physical characteristics, particularly meat color, are crucial for the consumer's perception of meat quality, as they influence its attractiveness and freshness. A study on broilers fed diets supplemented with 20% dried OPulp and/or OLeav reported no adverse effects on meat quality parameters such as pH, lightness (L*), redness (a*), cooking loss, and tenderness. However, the study observed a significant decrease in meat yellowness (b*), suggesting a potential impact on meat color [24]. However, findings

indicate that incorporating olive by-products into animal diets can influence meat color and other physical properties, potentially enhancing meat quality [24,25]. Furthermore, OPulp is an important source of unsaturated FA, such as oleic acid, palmitic acid, and linolenic acid, and can affect the FA profile of animal tissues [26]. Papadomichelakis et al. [24] found that the addition of 50 g/kg of dried OPulp improved the FA profile of breast meat by decreasing the intramuscular saturated FA (SFA) content while increasing oleic acid and total MUFAs, which are known for their beneficial effects on the cardiovascular system.

Despite some nutritional limitations, such as the high fiber content, olive by-products could be included in broiler diets due to their antioxidant properties, which are attributed to phenolic compounds and their desirable FA composition, mainly being rich in MUFAs. Therefore, the aim of the present study was to evaluate the effects of OLeav and OPulp supplementation at 5% and 10% on the oxidative status and broiler breast meat quality.

2. Materials and Methods

The study took place at the research facility of the Department of Animal Science of the Biotechnical faculty at the University of Ljubljana. All procedures followed international scientific and ethical standards and received approval from the Animal Ethics Committee of the Veterinary Administration of the Republic of Slovenia (U3440138/2017/5).

2.1. Birds and Dietary Treatments

A group of 120-day-old male Ross 308 broilers was placed in 10 pens with 12 birds each on deep litter. The pens, measuring 95 × 126 cm, were fitted with a plastic feeder and five nipple drinkers. Feed and fresh water were available without restriction throughout the study. The birds were reared under a standard lighting schedule (18 h of light and 6 h of darkness), with changes at the beginning and end of the experiment.

From day 1 to 21, all broilers were fed a commercial starter diet without OLeav or OPulp. On day 21, the birds were assigned to five experimental groups, with two replicate pens per treatment (ten broilers per pen). During this period, the animals were fed a finisher diet. The control group (Cont) received a complete finisher diet prepared following to the nutritional standards for Ross 308 broilers [27]. The other groups received a complete finisher diet supplemented with either 5% (OLEav5) and 10% (OLEav10) OLeav or 5% (OPulp5) and 10% (OPulp10) OPulp. The experiment lasted until the 42nd day.

The OLeav used in this study originated from organic olive groves in Slovenia provided by the Institute of Olive Culture. The leaves were dried at 45 °C using both air drying and a drying oven to ensure uniform dehydration while preserving their bioactive properties. The OPulp was provided by a local Slovenian olive oil producer in Koper. It consisted of olive cake without pits, as the olives were deboned prior to further processing. Like the OLeav, the OPulp was dried and ground before being incorporated into the experimental diets. The chemical composition and the mineral and polyphenols content of OLeav and OPulp were analyzed and are listed in Table 1.

Table 1. Chemical composition and evaluated energy of olive leaves and olive pulp.

Component	Olive Leaves (OLEav)	Olive Pulp (OPulp)
Chemical composition and metabolic energy		
Dry matter (g/kg)	848.4	976.1
Crude protein (g/kg)	72.4	91.9
Crude fat (g/kg)	20.2	226.0
Crude fiber (g/kg)	247.7	239.4
Crude ash (g/kg)	47.5	59.3
Nitrogen free extract (g/kg)	460.8	359.5
Evaluated metabolizable energy (MJ/kg)	3.2	8.5

Table 1. *Cont.*

Component	Olive Leaves (OLeav)	Olive Pulp (OPulp)
Mineral and polyphenols content		
K (g/kg)	6.78	26.7
P (g/kg)	0.89	2.18
Ca (g/kg)	13.43	2.70
Na (g/kg)	0.32	0.20
Mg (g/kg)	1.09	0.77
Fe (mg/kg)	99.2	155.8
Zn (mg/kg)	107.7	117.2
Polyphenols (g/kg)	66.9	29.0

The experimental finisher diets components and calculated nutrient contents are provided in Table 2.

Table 2. Nutritional composition (%) and calculated values of nutrients content of the finisher diets.

Component	Experimental Groups ¹				
	Cont	OLeav5	OLeav10	OPulp5	OPulp10
Composition, %					
Maize (%)	56.71	54.80	52.90	56.30	55.87
Wheat bran (%)	5.00	2.50	-	2.50	-
Alfalfa (%)	5.00	2.50	-	2.50	-
Soya bean meal (%)	26.00	27.00	28.00	26.60	27.20
Oil (%)	3.40	4.20	5.00	3.00	2.60
Salt (%)	0.44	0.44	0.44	0.44	0.44
Limestone (%)	0.99	0.95	0.90	1.04	1.10
Monocalcium phosphate (%)	1.45	1.55	1.65	1.55	1.65
L-lysine-HCl (%)	0.17	0.19	0.20	0.20	0.22
DL-methionine (%)	0.28	0.29	0.31	0.29	0.31
Threonine (%)	0.06	0.08	0.10	0.08	0.10
Tryptophan (%)	-	-	-	-	0.01
Premix (%) ^{2,3}	0.50	0.50	0.50	0.50	0.50
Olive leaves (%)	-	5.00	10.00	-	-
Olive pulp (%)	-	-	-	5.00	10.00
Calculated values					
Metabolizable energy (MJ/kg)	12.18	12.18	12.18	12.18	12.18
Digestible lysine (g/kg)	10.0	10.0	10.0	10.0	10.0
Available P (g/kg)	4.1	4.1	4.1	4.1	4.1

¹ Cont = no supplement; OLeav5 = 5% olive leaves; OLeav10 = 10% olive leaves; OPulp5 = 5% olive pulp; OPulp10 = 10% olive pulp. ² Feed mixtures also contain coccidiostat, Maxiban® G160 (Elanco Products Co., Hampshire, UK). ³ Calculated to meet the mineral and vitamin specifications for Ross 308 finisher diets (per kilogram): Cu 15 mg, I 1 mg, Fe 43 mg, Mn 101 mg, Se 0.3 mg, Zn 74 mg, vitamin A 10,000 IU, vitamin D3 5000 IU, vitamin E 50 IU, vitamin K 3 mg, thiamine (B1) 3 mg, riboflavin (B2) 6 mg, niacin (B3) 60 mg, pantothenic acid 15 mg, pyridoxine (B6) 4 mg, biotin 0.15 mg, folic acid 2 mg, vitamin B12 0.15 mg.

During the trial, feed samples were collected for chemical analysis and FA composition. Table 3 presents the proximate composition of the experimental finisher diets.

Table 3. Chemical composition, concentration of malondialdehyde (MDA), α -, γ + β -, and δ -tocopherol, polyphenol content, and fatty acid (FA) composition of the finisher diets.

Component	Experimental Groups				
	Cont	OLeav5	OLeav10	OPulp5	OPulp10
Chemical composition					
Dry matter (g/kg)	896	897	901	898	899
Crude protein (g/kg)	173	176	178	176	174

Table 3. Cont.

Component	Experimental Groups				
	Cont	OLeav5	OLeav10	OPulp5	OPulp10
Chemical composition					
Crude fat (g/kg)	60.3	70.2	77.0	65.9	74.3
Crude fiber (g/kg)	45.1	49.7	51.4	55.3	53.1
Crude ash (g/kg)	62.1	52.5	51.5	52.3	52.2
Ca (g/kg)	10.5	10.4	10.7	10.2	10.3
K (g/kg)	11.5	11.1	11.1	11.9	11.8
P (g/kg)	7.04	6.96	6.91	7.04	6.79
Na (g/kg)	4.98	2.12	2.41	2.24	2.18
Mg (g/kg)	2.05	1.91	1.83	1.89	1.68
MDA, tocopherol isomers and polyphenols					
MDA (nmol/g)	22.32	27.53	36.78	16.93	6.53
α -tocopherol (mg/kg)	54.90	59.03	66.50	53.99	49.39
γ + β -tocopherol (mg/kg)	48.64	51.02	56.84	44.96	40.56
δ -tocopherol (mg/kg)	11.72	10.41	10.42	9.28	6.08
Polyphenols (g/kg)	1.82	4.74	5.67	2.46	1.98
Fatty acids (FAs) composition ¹ (g of FA/100 g FAs)					
C16:0	11.96	11.86	11.72	12.24	12.42
C16:1 n – 7	0.14	0.13	0.13	0.29	0.44
C18:0	2.59	2.83	2.99	2.70	2.62
Σ C18: 1 ²	21.28	21.87	22.12	30.00	37.02
C18:2 n – 6	56.06	54.94	54.27	48.06	41.73
C18:3 n – 3	7.10	7.42	7.77	5.78	4.83
Sum of SFA	15.22	15.43	15.51	15.64	15.73
Sum of MUFA	21.61	22.21	22.44	30.52	37.70
Sum of PUFA	63.17	62.37	62.04	53.84	46.57
n – 3 PUFA	7.10	7.42	7.77	5.78	4.83
n – 6 PUFA	56.06	54.94	54.27	48.06	41.73
n – 6/n – 3 PUFA	7.90	7.40	6.98	8.33	8.63

The experimental groups are labeled in Table 2. MDA = malondialdehyde; SFA = saturated FAs; MUFA = monounsaturated FAs; PUFA = polyunsaturated FAs. ¹ Values represent means of two analyses per sample. Only dietary important FAs are listed. ² Sum of isomers.

2.2. Experimental Procedure and Sample Collection

On day 21, the chickens were individually marked with metal leg bands. The body weight (BW) of the chickens was recorded weekly from day 21 until the end of the trial. Growth performance parameters, including BWG, average daily feed intake (ADFI), and feed conversion ratio (FCR), were monitored throughout the trial.

At the end of the experimental period, 10 birds were randomly chosen from each group. Broilers were weighed and humanely sacrificed following standard procedures. They were stunned using electrical shears and then bled to ensure complete exsanguination. Blood and breast muscle samples were collected for further analysis. Plasma samples were collected in K₂-EDTA tubes for analysis of MDA and vitamin E concentrations. Serum samples for determination of antioxidant capacity of lipid- (ACL) and water-soluble (ACW) compounds and activity of gamma-glutamyl transferase (GGT), aspartate aminotransferase (AST), and alanine aminotransferase (ALT) were collected in tubes with clot activator. Plasma and serum samples were then transferred to 1.5 mL Eppendorf tubes and stored at –80 °C prior to further analysis. Breast muscles (*Pectoralis major*) were collected after 24 h of cooling at 4 °C.

2.3. Chemical Analyses

2.3.1. Chemical Analysis of Olive Leaves, Pulp and Feeds

The samples taken, both additives and all experimental feed mixtures, were homogenized and analyzed. The concentrations of MDA and vitamin E as well as the FA composition of the experimental diets were determined according to the methods described by Voljč et al. [5]. The polyphenol content was analyzed according to the method of Hinojosa-Nogueira et al. [28].

2.3.2. Analysis of Vitamin and Antioxidant Concentrations

The levels of α - and γ -tocopherol (vitamin E) and the degree of lipid oxidation (MDA concentration) in homogenized feed, plasma, and breast muscle samples were assessed using an Agilent 1260 Infinity high performance liquid chromatograph (HPLC) (Santa Clara, CA, USA) following the methods outlined by Voljč et al. [5] and Pečjak et al. [29]. The concentrations of ACW and ACL in blood serum and breast muscle samples were determined using PhotoChem (Analytik Jena, Jena, Germany) following the ACW-Kit and ACL-Kit protocols provided by the manufacturer, as described by Pečjak Pal et al. [29].

2.3.3. Analysis of Liver Enzyme Activities and Fatty Acid Profiles

Liver enzymes (GGT, AST, and ALT) activities in serum were measured using Randox kits (Laboratories Ltd., Ardmore, Crumlin Co., Crumlin, UK) and the RX Daytona biochemical autoanalyzer (Randox, Crumlin Co., Crumlin, UK). The FA profile of the feed and breast muscle samples was analyzed using a gas chromatograph (Agilent 6890, Santa Clara, CA, USA) equipped with an Omegawax 320 column (30 m $\times$ 0.32 mm i.d. $\times$ 0.25 μ m; Supelco, Bellefonte, PA, USA) and an FID detector, as described by Rezar et al. [30].

2.3.4. Analysis of Meat Quality and Oxidation Stability

The left side of the muscle was chopped into evenly distributed slices, packed in zippered polypropylene plastic bags and stored at -80°C . At 24 h postmortem, the electrical conductivity of the breast muscle was measured using the LF-Star device (Ingenieurbüro Matthäus, Nobitz, Germany), the pH of the breast muscle was measured using a portable pH meter (Mettler Toledo Seven2Go, Mettler Toledo, Schwerzenbach, Switzerland), while breast meat color was assessed using a Minolta CR 300 colorimeter (Minolta Co., Ltd., Osaka, Japan) as described by Voljč et al. [5]. Color was expressed as CIE lightness (L^*), redness (a^*), and yellowness (b^*). Drip loss was measured 48 h postmortem using a bag method as described by Honikel [31]. Prior to analysis, liquid nitrogen and a knife mill (Grindomix GM200, Retsch GmbH and Co., Haan, Germany) were used to homogenize all breast meat samples. The total fat content in the chicken breast muscle was determined using the Soxhlet extraction with acid hydrolysis method as described by Pérez-Palacios et al. [32]. FA profiles of breast meat were determined according to the methods described by Voljč et al. [5].

2.4. Statistical Analyses

The data were analyzed with the GLM procedure of the SAS software (Ver. 9.4, SAS Institute, Cary, NC, USA). A one-way ANOVA was conducted for all parameters, with the experimental group set as a fixed effect. Differences between groups were assessed using Tukey's multiple comparison test. In the Section 3, the least square means and the standard error of the mean (SEM) as a dispersion parameter are shown. Statistical significance was assumed at $p < 0.05$.

3. Results

3.1. Growth Performance

During the experimental period (days 21–42), the chickens adapted well to the experimental conditions, and no health issues were observed. No significant differences ($p > 0.05$) were observed regarding the final BW, BWG, or ADFI. However, broilers fed 10% OLeav or OPulp tended to have a higher FCR than those fed 5% OLeav or OPulp or than the control group (Table 4), although the differences were not statistically significant.

Table 4. Effects of olive leaves and olive pulp supplementation on the growth performance of broiler chickens during the experimental period (n = 10/group).

Parameter	Experimental Groups					SEM	p-Value
	Cont	OLeav5	OLeav10	OPulp5	OPulp10		
BW d21 (kg)	0.91	0.83	0.84	0.85	0.92	31.9	0.158
BW d42 (kg)	2.77	2.53	2.64	2.66	2.62	81.6	0.362
BWG d21–42 (kg)	1.87	1.75	1.80	1.81	1.74	64.9	0.619
ADFI d21–42 (g/day)	145.3	141.2	164.8	146.5	148.9	5.8	0.175
FCR d21–42 (g/g) ¹	1.83	1.94	2.22	1.82	2.08	0.12	0.214

Nomenclature of experimental groups as in Table 1. BW = body weight; BWG = body weight gain; ADFI = average daily feed intake; FCR = feed conversion ratio. ¹ Calculated by the ratio of total feed intake to total BWG.

3.2. MDA Levels, Antioxidants and Activity of Liver Enzymes in the Blood

The experimental groups showed no significant differences in plasmatic MDA concentration and serum ACL levels. There were also no differences in liver enzyme activities (GGT, AST, ALT) between the groups (Table 5). However, birds in group OLeav10 had a 24% lower plasma α -tocopherol content than those in group OLeav5. Similarly, broilers fed diets supplemented with 10% OLeav or OPulp had lower plasma γ -tocopherol levels than those 5% OLeav or OPulp. No differences were observed in plasma α - and γ -tocopherol levels between the Cont group and the supplemented groups. In addition, supplementation with 5% OLeav or OPulp and 10% OLeav reduced serum ACW levels by 16%, 28%, and 28%, respectively, compared to the Cont group.

Table 5. Effect of dietary inclusion of olive leaves and olive pulp on plasma MDA, α - and γ -tocopherol, serum ACW and ACL, and serum liver enzymes activity in broiler chickens.

Parameter	Experimental Groups					SEM	p-Value
	Cont	OLeav5	OLeav10	OPulp5	OPulp10		
Plasma MDA (nmol/mL)	0.31	0.30	0.32	0.33	0.25	0.023	0.092
Plasma α -tocopherol (μ g/mL)	0.115 ^{ab}	0.136 ^a	0.103 ^b	0.133 ^{ab}	0.120 ^{ab}	0.76	0.027
Plasma γ -tocopherol (μ g/mL)	0.012 ^{ab}	0.016 ^a	0.009 ^b	0.014 ^a	0.009 ^b	0.09	<0.0001
Serum ACW (nmol/mL)	310.2 ^a	261.8 ^b	224.7 ^b	224.7 ^b	362.9 ^{ab}	25.03	0.002
Serum ACL (nmol/mL)	307.3	397.1	368.7	338.8	335.7	35.19	0.447
Liver enzymes							
Serum GGT (U/L)	20.95	24.26	24.65	22.92	23.49	1.74	0.558
Serum AST (U/L)	248.6	261.9	277.8	257.7	261.6	21.36	0.869
Serum ALT (U/L)	3.47	2.38	3.10	3.96	3.30	0.59	0.445

The experimental groups are labeled in Table 2. MDA = malondialdehyde; ACW = antioxidant capacity of water-soluble antioxidants; ACL = antioxidant capacity of lipid-soluble antioxidants; GGT = gamma-glutamyl transferase.; AST = aspartate aminotransferase; ALT = alanine aminotransferase. ^{ab} Superscript letters that differ within the row indicate significant differences ($p < 0.05$).

3.3. MDA Levels and Antioxidants in Breast Muscle

As shown in Table 6, the addition of OLeav or OPulp had no effect on the meat quality parameters, although the lightness of the breast meat decreased in the OLeav10 group compared to the control group. Dietary inclusion of 5% OPulp increased the α -tocopherol content in the breast muscle by 19%, 34%, and 19% compared to supplementation with 5% OLeav, 10% OLeav, and 10% OPulp, respectively. In addition, supplementation with 10% OLeav reduced the α -tocopherol content in the breast muscle by 9% compared to the Cont group. Similarly, broilers fed 10% OLeav or 10% OPulp had a lower γ -tocopherol content in the breast muscle than those fed 5% OPulp. In addition, the γ -tocopherol content in the breast muscle of the OPulp10 group was significantly lower compared to the Cont group. In contrast, there were no significant differences in the MDA or the ACW and ACL levels between the experimental groups.

Table 6. Effect of dietary inclusion of olive leaves and olive pulp on the meat quality and oxidative stability of breast muscle in broiler chickens.

Parameter	Experimental Groups					SEM	p-Value
	Cont	OLeav5	OLeav10	OPulp5	OPulp10		
Meat quality traits							
Electrical conductivity (S/m)	3.52	3.53	3.84	4.20	3.92	0.28	0.386
pH _{24h}	5.95	6.11	6.05	5.94	6.04	0.04	0.500
Drip loss (%)	2.15	1.74	2.12	1.85	1.49	0.26	0.368
L*—lightness	54.00 ^a	49.25 ^{ab}	48.03 ^b	53.05 ^{ab}	52.90 ^{ab}	1.48	0.023
a*—redness	0.49	0.95	0.83	0.87	0.90	0.25	0.711
b*—yellowness	11.79	11.47	11.89	10.55	10.30	0.62	0.250
Oxidative stability							
MDA (nmol/g)	0.99	0.66	0.89	0.50	1.16	0.18	0.091
α-tocopherol (μg/100 g)	536.78 ^{ab}	491.18 ^{bc}	437.03 ^c	584.64 ^a	489.50 ^{bc}	22.19	0.001
γ-tocopherol (μg/100 g)	86.57 ^{ab}	77.69 ^{abc}	69.92 ^{bc}	93.99 ^a	62.76 ^c	5.13	0.001
ACW (nmol/g)	170.38	182.63	166.95	165.19	185.63	9.49	0.437
ACL (nmol/g)	6.43	6.27	6.42	7.54	6.55	0.42	0.232

The experimental groups are labeled in Table 2. MDA = malondialdehyde; ACW = antioxidant capacity of water-soluble antioxidants; ACL = antioxidant capacity of lipid-soluble antioxidants; n = 10 replicants/treatment; ^{abc} Superscript letters that differ within the row indicate significant differences ($p < 0.05$).

3.4. Fatty Acid Composition of Breast Muscle

The FA profile of the fresh breast meat is shown in Table 7. Broilers fed 5% OLeav had a lower total fat content as well as a lower SFAs and MUFAs content compared to the Cont and OPulp5 groups. In addition, supplementation with 10% OLeav resulted in higher levels of PUFAs, including n – 3 and n – 6 PUFAs, in breast meat compared to the OLeav5 group. The ratio of n – 6 to n – 3 PUFAs was narrower in the breast meat of chickens fed OLeav than in those fed OPulp.

Table 7. Effect of experimental diets on total fat content and fatty acid profile in breast muscle.

Parameter	Experimental Groups					SEM	p-Value
	Cont	OLeav5	OLeav10	OPulp5	OPulp10		
Fat content (g/100 g)	1.31 ^b	1.10 ^b	1.59 ^{ab}	1.62 ^a	1.50 ^{ab}	0.13	0.023
Fatty acids (FAs) ¹ (mg FAs/ 100 g of total FAs)							
Σ C16:1 ²	57.74 ^a	22.88 ^b	34.72 ^{ab}	48.82 ^a	40.65 ^{ab}	5.69	0.002
C18:0	114.61	89.58	110.65	107.25	105.82	6.23	0.076
Σ C18:1 ²	417.02 ^a	231.53 ^b	349.27 ^{ab}	440.80 ^a	455.62 ^a	39.31	0.002

Table 7. Cont.

Parameter	Experimental Groups					SEM	p-Value
	Cont	OLeav5	OLeav10	OPulp5	OPulp10		
Fatty acids (FAs) ¹ (mg FAs/ 100 g of total FAs)							
C18:2 n – 6	439.99 ^{ab}	291.43 ^b	487.48 ^a	398.94 ^{ab}	332.21 ^{ab}	41.35	0.015
C18:3 n – 3	37.48 ^{ab}	28.13 ^a	54.16 ^b	39.27 ^{ab}	29.83 ^a	5.54	0.009
C18:4 n – 3	0.99 ^a	0.37 ^b	0.88 ^a	0.90 ^a	0.73 ^{ab}	0.11	0.003
C20:2 n – 6	9.53 ^a	9.23 ^a	9.90 ^a	8.48 ^{ab}	6.48 ^b	0.54	0.001
C20:3 n – 6	11.86	9.85	10.39	11.12	10.34	0.49	0.061
C20:0 n – 6	64.99	65.12	71.15	73.41	72.64	2.23	0.021
C20:3 n – 3	2.14 ^{ab}	2.17 ^{ab}	2.38 ^a	1.83 ^b	1.31 ^c	0.12	<0.0001
C20:5 n – 3	3.64	3.29	3.45	3.66	3.31	0.22	0.611
C22:4 n – 6	15.07	14.66	13.98	15.72	15.71	0.47	0.065
C22:5 n – 3	13.71 ^a	17.01 ^{cb}	17.63 ^c	14.79 ^{ab}	14.82 ^{ab}	0.62	0.0004
C22:6 n – 3	9.09 ^a	10.54 ^{ab}	12.12 ^b	10.22 ^{ab}	11.71 ^{ab}	0.69	0.031
ΣSFA	480.56 ^a	325.21 ^b	429.57 ^{ab}	454.42 ^a	419.33 ^{ab}	29.55	0.010
ΣMUFA	480.53 ^a	258.54 ^b	387.52 ^{ab}	494.73 ^a	501.56 ^a	44.82	0.002
ΣPUFA	619.26 ^{ab}	452.33 ^b	683.95 ^a	578.67 ^{ab}	499.31 ^{ab}	48.22	0.015
Σn – 3 PUFA	77.83 ^{ab}	62.03 ^b	91.04 ^a	71.00 ^{ab}	61.94 ^b	5.93	0.008
Σn – 6 PUFA	541.43 ^{ab}	390.30 ^b	592.90 ^a	507.67 ^{ab}	437.37 ^{ab}	42.40	0.016
n – 6/n – 3 PUFA	6.94 ^{ab}	6.30 ^c	6.52 ^{bc}	7.14 ^a	7.02 ^a	0.12	<0.0001

The experimental groups are labeled in Table 2. SFA = saturated FAs; MUFA = monounsaturated FAs; PUFA = polyunsaturated FAs. ¹ Values represent means of two analyses per sample. Only dietary important FAs are listed. The sum of SFAs, MUFAs, and PUFAs are calculated from all analyzed FAs. ² Sum of isomers. ^{abc} Superscript letters that differ within the row indicate significant differences ($p < 0.05$).

4. Discussion

The integration of olive oil by-products into the diet of monogastric animals offers a sustainable approach to diminish environmental pollution while providing essential nutrients and bioactive compounds. In particular, OLeav and OPulp, derived from the extraction of olive oil, are rich in polyphenols, unsaturated FAs, and essential minerals, which have been associated with improved immunity, oxidative status, and productivity in poultry [26]. However, due to variations in nutritional value, fiber content, and digestibility [15], the optimal inclusion levels of olive oil by-products needs to be determined to balance the nutritional benefits with the potential limitations.

The analysis of the experimental feed mixtures used in this study showed that the experimental feed with 10% OLeav had the highest content of MDA, vitamin E, and polyphenols. This was to be expected as OLeav naturally has a higher polyphenol content than OPulp. The results of our study showed that OLeav contained 66.9 g/kg of polyphenols compared to 29.0 g/kg in Opulp, which is consistent with previous results [26]. Additionally, OLeav contains a higher proportion of SFA and PUFA, while Opulp is rich in MUFA [33], which was also partially confirmed by our results. This composition has a significant impact on oxidative stability and antioxidant capacity, which could improve broiler health and meat quality.

4.1. Growth Performance

Supplementation with OLeav and OPulp had no effect on the performance parameters of the broilers. However, due to the small number of broilers included in the present trial, the growth performance results should be interpreted as indicative of the trial conditions. Furthermore, our results are consistent with previous studies indicating that the addition of olive by-products in similar amounts to broiler diets can be conducted without adverse effects on growth performance [8]. For example, Al-Harhi [7] reported that the addition of dried olive cake at 5% and 10% to broiler diets during the growing period had no

significant effects on growth, feed intake, and FCR. The same trend was confirmed by Tufarelli et al. [34] and Dedousi et al. [35], who found that the addition of dried OPulp up to 10% in broiler diets had no detrimental effect on feed efficiency. In contrast, some studies reported that even lower levels of OLeav and OPulp impaired ADFI of broilers, probably due to the effect of OLeav on the bitter taste and feed palatability [22,24], while other studies reported improved growth indices in broilers supplemented with olive by-products [8]. To confirm the performance results of the present study, which indicate that both OLeav and OPulp can be included in broiler diets up to 10% without negatively affecting final BW, BWG, and feed efficiency, further studies with a larger number of birds are recommended.

4.2. MDA Levels, Antioxidants and Activity of Liver Enzymes in the Blood

Olive by-products have antioxidant properties that reduce the excessive formation of reactive oxygen species (ROS) and protect cells from oxidative damage [8]. In the present study, no significant differences were found between the experimental groups in plasma MDA levels, serum ACL levels, or liver enzyme activities, suggesting that dietary supplementation with OLeav and OPulp of up to 10% had no positive effects on the oxidative status or liver function of broilers. However, a reduction in plasma α -tocopherol concentrations was observed in broilers receiving 10% OLeav, while plasma γ -tocopherol levels were lower in both the OLeav10 and OPulp10 groups. Serum ACW levels were also significantly lower in birds in the OLeav5, OPulp5, and OLeav10 groups than in the control group. The observed reduction in antioxidant levels could be due to the polyphenol content in olive oil by-products, which, despite their antioxidant properties, can act as pro-oxidants under certain conditions, leading to a depletion of endogenous antioxidants [36]. On the other hand, previous studies reported a sparing effect of polyphenols on tocopherol concentrations in plasma of broilers [37] and their role in increasing antioxidant activities by preventing oxidation of low molecular weight antioxidants (e.g., ascorbate, tocopherols) [38]. In contrast to our results, previous studies have reported an improvement in antioxidant capacity and protection against oxidative damage after olive leaf extract supplementation, as reflected by lower ALT and ALP activity and lower MDA levels [22,39]. In addition, Gerasopoulos et al. [23] reported that olive oil mill wastewater supplementation improved the oxidative status of broilers, as evidenced by a significant increase in total antioxidant capacity (TAS) and a decrease in protein carbonyl and thiobarbituric acid reactive substances (TBARS). However, Elbaz et al. [40] reported increased MDA levels in broilers fed 10% and 15% OPulp, indicating increased oxidation of LDL.

4.3. MDA Levels and Antioxidants in Breast Muscle

Studies on the effects of OPulp and OLeav supplementation on meat quality parameters such as pH, color, tenderness and lipid oxidation are still limited despite their importance for consumer preference. The available results suggest that the addition of olive by-products to animal diets can significantly alter meat color and other physical characteristics, potentially enhancing overall meat quality and influencing consumer perception [24,25]. Shafey et al. [19] reported that the replacement of 15 or 30 g/kg wheat bran with OLeav had no significant effect on carcass characteristics of chickens, while the addition of 50 g/kg OLeav reduced both liver and carcass weight. On the other hand, Almuhayawi et al. [41] reported that supplementation with 75 to 150 μ g/kg OLeav powder significantly reduced abdominal fat and increased breast muscle weight compared to the control group. The present study shows that the inclusion of OLeav and OPulp to broiler diets can affect the α -tocopherol concentration in breast meat and meat brightness after 24 h. However, the results regarding lipid oxidation are not conclusive, as the MDA levels observed in this study were highly variable between the groups (between 0.50 and 1.16 nmol/g), with

no significant statistical differences. This heterogeneity suggests that although olive by-products contain bioactive compounds with antioxidant potential, their effect on oxidative stability may be influenced by several factors, including dosage, diet composition, and individual metabolic differences between chickens. In addition, a study on slow-growing broilers showed that supplementing the diet with up to 10% dried OPulp improved the oxidative stability of the meat, primarily due to the high polyphenol content, which acts as a natural antioxidant by reducing lipid peroxidation [34]. Further studies are needed to clarify the relationship between olive by-product supplementation and oxidative stability, considering the variability of response observed in this study.

4.4. Fatty Acid Composition of Breast Muscle

Dietary OLeav and OPulp also affects the FA profile of broiler breast muscle. The results of this study showed that supplementation with 5% OLeav resulted in lower levels of SFAs and MUFAs in the breast meat than in the control group. In addition, broilers receiving 10% OLeav had a higher content of docosahexaenoic acid (C22:6 n – 3) than the control group. Studies have also shown that the inclusion of olive pomace oil-derived sour oil as a fat source in poultry feed results in meat enriched in MUFAs without compromising oxidative stability or consumer acceptability. This change in FA composition may contribute to the production of healthier meat products [42]. Moreover, Papadomichelakis et al. [24] reported that inclusion of dried olive pulp, an oleic acid rich residue, to broiler finisher diets increased intramuscular oleic acid and total MUFA in proportion to the inclusion level. Similarly, replacing part of the maize content with olive cake meal in the broiler diet resulted in a shift in the FA profile in the breast muscle towards higher unsaturation, with oleic acid (C18:1 n – 9) content increasing significantly with increasing olive cake content (from ~40 to 45 mg/100 g fat), while the palmitic acid (C16:0) content in the meat decreased with higher inclusion. In addition, the intake of olive cake meal increased the content of PUFA such as linoleic acid (C18:2 n – 6) and α -linolenic acid (C18:3 n – 3) in breast muscle [43].

5. Conclusions

Based on the results of this study, dietary inclusion of olive leaves and pulp had no significant effect on growth performance, liver function, or MDA levels in the blood and breast meat of broilers. However, the inclusion of 10% OLeav reduced the α -tocopherol levels in plasma and muscle as well as the ACW content in serum, indicating an influence on the oxidative balance. The addition of both olive by-products to broiler diets influenced the FA composition of the breast muscle without compromising meat quality parameters. These results suggest that supplementation with up to 5% olive leaves or pulp can be included in broiler diets without significant effects on oxidative status or meat quality, possibly due to the level of addition. Further studies with varying inclusion levels are needed to clarify the relationship between olive by-product supplementation and oxidative stability. Additionally, from an economic and environmental perspective, further studies under commercial production conditions are recommended, addressing the practical limitations related to the drying costs of olive by-products, which are particularly relevant for olive pulp processing.

Author Contributions: Conceptualization, V.R., J.S. and A.L.; formulation of experimental diets: V.R. and J.S.; methodology, investigation: A.L., J.S. and V.R.; formal analysis: V.R. and A.L.; data curation: V.R., M.P.P. and A.L.; writing—original draft preparation: V.R. and M.P.P.; supervision, A.L. All authors have read and agreed to the published version of the manuscript.

Funding: The research was funded by the Slovenian Research and Innovation Agency (Ljubljana, Slovenia), grant number: P4-0097 and the Slovenian Research and Innovation Agency and the Ministry of Agriculture, Forestry and Food: CRP V4-1621.

Institutional Review Board Statement: All procedures with animals were conducted in accordance with the prevailing legislation on animal experimentation in Slovenia, which aligns with the EU regulations regarding research on experimental animals. The study protocol was approved by the Animal Ethics Committee of the Veterinary Administration of the Republic of Slovenia (project licence number U3440138/2017/5). The experiment was undertaken in the research facility of the Department of Animal Science, Biotechnical Faculty of University of Ljubljana, Slovenia.

Data Availability Statement: The original contributions presented in the study are included in the article; further inquiries can be directed to the corresponding author.

Acknowledgments: The authors are grateful to the Science and Research Centre Koper, its institutes and infrastructure units, and the Institute for Oliveculture, the lead partner of the CRP V4-1621 project. In particular, we would like to thank Bojan Butinar, Milena Bučar Miklavčič and Maja Podgornik for providing olive leaves and pulp and for performing the laboratory analyzes.

Conflicts of Interest: The authors declare no conflicts of interest. The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

Abbreviations

The following abbreviations are used in this manuscript:

OPulp	Olive pulp
OLeav	Olive leaves
MDA	Malondialdehyde
ACL	Antioxidant capacity of lipid soluble compounds
ACW	Antioxidant capacity of water soluble compounds
FA	Fatty acids
MUFAs	Monounsaturated fatty acids
LDL	Low-density lipoproteins
BWG	Body weight gain
NSP	Non-starch polysaccharides
SFA	Saturated fatty acid
ADFI	Average daily feed intake
FCR	Feed conversion ratio
GGT	Gamma-glutamyl transferase
AST	Aspartate aminotransferase
ALT	Alanine aminotransferase
HPLC	High performance liquid chromatography
BW	Body weight
ROS	Reactive oxygen species
PUFAs	Polyunsaturated fatty acids
TAS	Total antioxidant capacity
TBARS	Thiobarbituric acid reactive substances
LDL	Low-density lipoproteins

References

1. Surai, P.F.; Kochish, I.I.; Fisinin, V.I.; Kidd, M.T. Antioxidant defence systems and oxidative stress in poultry biology: An update. *Antioxidants* **2019**, *8*, 235. [[CrossRef](#)] [[PubMed](#)]
2. Estévez, M. Oxidative damage to poultry: From farm to fork. *Poult. Sci.* **2015**, *94*, 1368–1378. [[CrossRef](#)] [[PubMed](#)]
3. Jena, B.P.; Panda, N.; Patra, R.C.; Mishra, P.K.; Behura, N.C.; Panigrahi, B. Supplementation of vitamin E and C reduces oxidative stress in broiler breeder hens during summer. *Food Nutr. Sci.* **2013**, *4*, 33–37. [[CrossRef](#)]

4. Leskovec, J.; Levart, A.; Perić, L.; Đukić Stojčić, M.; Tomović, V.; Pirman, T.; Salobir, J.; Rezar, V. Antioxidative effects of supplementing linseed oil-enriched diets with α -tocopherol, ascorbic acid, selenium, or their combination on carcass and meat quality in broilers. *Poult. Sci.* **2019**, *98*, 6733–6741. [[CrossRef](#)] [[PubMed](#)]
5. Voljč, M.; Frankič, T.; Levart, A.; Nemec, M.; Salobir, J. Evaluation of different vitamin E recommendations and bioactivity of α -tocopherol isomers in broiler nutrition by measuring oxidative stress in vivo and the oxidative stability of meat. *Poult. Sci.* **2011**, *90*, 1478–1488. [[CrossRef](#)] [[PubMed](#)]
6. Salami, S.A.; Majoka, M.A.; Saha, S.; Garber, A.; Gabarrou, J.-F. Efficacy of dietary antioxidants on broiler oxidative stress, performance and meat quality: Science and market. *Avian Biol. Res.* **2015**, *8*, 65–78. [[CrossRef](#)]
7. Al-Harathi, M.A. The efficacy of using olive cake as a by-product in broiler feeding with or without yeast. *Ital. J. Anim. Sci.* **2016**, *15*, 512–520. [[CrossRef](#)]
8. Mahasneh, Z.M.H.; Abdelnour, S.; Ebrahim, A.; Almasodi, A.G.S.; Moustafa, M.; Alshaharni, M.O.; Algopish, U.; Tellez-Isaias, G.; Abd El-Hack, M.E. Olive oil and its derivatives for promoting performance, health, and struggling thermal stress effects on broilers. *Poult. Sci.* **2024**, *103*, 103348. [[CrossRef](#)]
9. Nunes, M.A.; Pimentel, F.B.; Costa, A.S.G.; Alves, R.C.; Oliveira, M.B.P.P. Olive by-products for functional and food applications: Challenging opportunities to face environmental constraints. *Innov. Food Sci. Emerg. Technol.* **2016**, *35*, 139–148. [[CrossRef](#)]
10. Abbattista, R.; Ventura, G.; Calvano, C.D.; Cataldi, T.R.I.; Losito, I. Bioactive Compounds in waste by-products from olive oil production: Applications and structural characterization by mass spectrometry techniques. *Foods* **2021**, *10*, 1236. [[CrossRef](#)]
11. Tzamaloukas, O.; Neofytou, M.C.; Simitzis, P.E. Application of olive by-products in livestock with emphasis on small ruminants: Implications on rumen function, growth performance, milk and meat quality. *Animals* **2021**, *11*, 531. [[CrossRef](#)] [[PubMed](#)]
12. Yenigeri, M.; Filik, A.G.; Filik, G. The effect of olive wastes for poultry feed on growth performance of broilers: A review. *J. Glob. Innov. Agric. Sci.* **2021**, *9*, 163–166. [[CrossRef](#)]
13. Paiva-Martins, F.; Barbosa, S.; Silva, M.; Monteiro, D.; Pinheiro, V.; Mourão, J.L.; Fernandes, J.; Rocha, S.; Belo, L.; Santos-Silva, A. The effect of olive leaf supplementation on the constituents of blood and oxidative stability of red blood cells. *J. Funct. Foods* **2014**, *9*, 271–279. [[CrossRef](#)]
14. Panel, E.; Nda, A. Scientific opinion on the substantiation of health claims related to polyphenols in olive and protection of LDL particles from oxidative damage (ID 1333, 1638, 1639, 1696, 2865), maintenance of normal blood HDL cholesterol concentrations (ID 1639), mainte. *EFSA J.* **2011**, *9*, 1–25. [[CrossRef](#)]
15. Sayehban, P.; Seidavi, A.; Dadashbeiki, M.; Ghorbani, A.; de Araújo, W.A.G.; Durazzo, A.; Lucarini, M.; Gabrielli, P.; Omri, B.; Albino, L.F.T.; et al. Olive pulp and exogenous enzymes feed supplementation effect on the carcass and offal in broilers: A preliminary study. *Agriculture* **2020**, *10*, 359. [[CrossRef](#)]
16. Omar, M.A.; Ottoman, R.A.; Abu Baker, B.M.; Zaazaa, A. Response of broiler chicks to a high olive pulp diet supplemented with two antibiotics. *Dirasat Agric. Sci.* **2003**, *30*, 137–142.
17. de Oliveira, C.O.; Roll, A.A.P.; Medeiros Gonçalves, F.M.; Lopes, D.C.N.; Xavier, E.G. Olive pomace for the feeding of commercial poultry: Effects on performance, meat and eggs quality, haematological parameters, microbiota and immunity. *Worlds Poult. Sci. J.* **2021**, *77*, 363–376. [[CrossRef](#)]
18. Sateri, S.; Seidavi, A.; Bouyeh, M.; Neumann, P.; Kutzler, M.; Laudadio, V.; Loperfido, F.; Tufarelli, V. Effect of olive meal and supplemental enzymes on performance traits, blood biochemistry, humoral immunity response and caecal microbiota of broilers. *S. Afr. J. Anim. Sci.* **2017**, *47*, 804–812. [[CrossRef](#)]
19. Shafey, T.M.; Al-Ruqaei, I.M.; Almufarij, S.I. Effect of feeding olive leaves extract (oleuropein) on the performance, nutrient utilization, small intestine and carcass characteristics of broiler chickens. *J. Anim. Vet. Adv.* **2013**, *12*, 740–746. [[CrossRef](#)]
20. Sayehban, P.; Seidavi, A.; Dadashbeiki, M.; Ghorbani, A.; Araújo, W.A.G.; Albino, L.F.T. Effects of different levels of two types of olive pulp with or without exogenous enzyme supplementation on broiler performance and economic parameters. *Rev. Bras. Cienc. Avic.* **2016**, *18*, 489–500. [[CrossRef](#)]
21. Pirman, T.; Rezar, V.; Vrecl, M.; Salobir, J.; Levart, A. Effect of olive leaves or marigold petal extract on oxidative stress, gut fermentative activity, and mucosa morphology in broiler chickens fed a diet rich in n-3 polyunsaturated fats. *J. Poult. Sci.* **2021**, *58*, 119–130. [[CrossRef](#)] [[PubMed](#)]
22. Xie, P.; Deng, Y.; Huang, L.; Zhang, C. Effect of olive leaf (*Olea europaea* L.) extract addition to broiler diets on the growth performance, breast meat quality, antioxidant capacity and caecal bacterial populations. *Ital. J. Anim. Sci.* **2022**, *21*, 1246–1258. [[CrossRef](#)]
23. Gerasopoulos, K.; Stagos, D.; Kokkas, S.; Petrotos, K.; Kantas, D.; Goulas, P.; Kouretas, D. Feed supplemented with byproducts from olive oil mill wastewater processing increases antioxidant capacity in broiler chickens. *Food Chem. Toxicol.* **2015**, *82*, 42–49. [[CrossRef](#)] [[PubMed](#)]
24. Papadomichelakis, G.; Pappas, A.C.; Tsiplakou, E.; Symeon, G.K.; Sotirakoglou, K.; Mpekis, V.; Fegeros, K.; Zervas, G. Effects of dietary dried olive pulp inclusion on growth performance and meat quality of broiler chickens. *Livest. Sci.* **2019**, *221*, 115–122. [[CrossRef](#)]

25. Vasilopoulou, K.; Papadopoulos, G.A.; Lioliopoulou, S.; Pyrka, I.; Nenadis, N.; Savvidou, S.; Symeon, G.; Dotas, V.; Panitsidis, I.; Arsenos, G.; et al. Effects of dietary supplementation of a resin-purified aqueous-isopropanol olive leaf extract on meat and liver antioxidant parameters in broilers. *Antioxidants* **2023**, *12*, 1723. [CrossRef]
26. Ferlisi, F.; Tang, J.; Cappelli, K.; Trabalza-Marinucci, M. Dietary supplementation with olive oil co-products rich in polyphenols: A novel nutraceutical approach in monogastric animal nutrition. *Front. Vet. Sci.* **2023**, *10*, 1272274. [CrossRef]
27. Aviagen Ross 308 Broiler: Nutrition Specifications. 2022. Available online: https://aviagen.com/assets/Tech_Center/Ross_Broiler/Ross-BroilerNutritionSpecifications2022-EN.pdf (accessed on 15 November 2024).
28. Hinojosa-Nogueira, D.; Muros, J.; Rufián-Henares, J.A.; Pastoriza, S. New method to estimate total polyphenol excretion: Comparison of fast blue bb versus folin-ciocalteu performance in urine. *J. Agric. Food Chem.* **2017**, *65*, 4216–4222. [CrossRef] [PubMed]
29. Pečjak Pal, M.; Leskovec, J.; Levart, A.; Pirman, T.; Salobir, J.; Rezar, V. Comparison of high n-3 PUFA levels and cyclic heat stress effects on carcass characteristics, meat quality, and oxidative stability of breast meat of broilers fed low-and high-antioxidant diets. *Animals* **2024**, *14*, 3314. [CrossRef] [PubMed]
30. Rezar, V.; Salobir, J.; Levart, A.; Tomažin, U.; Škrlep, M.; Batorek Lukač, N.; Čandek-Potokar, M. Supplementing entire male pig diet with hydrolysable tannins: Effect on carcass traits, meat quality and oxidative stability. *Meat Sci.* **2017**, *133*, 95–102. [CrossRef]
31. Honikel, K.O. Reference methods for the assessment of physical characteristics of meat. *Meat Sci.* **1998**, *49*, 447–457. [CrossRef]
32. Pérez-Palacios, T.; Ruiz, J.; Martín, D.; Muriel, E.; Antequera, T. Comparison of different methods for total lipid quantification in meat and meat products. *Food Chem.* **2008**, *110*, 1025–1029. [CrossRef] [PubMed]
33. Hannachi, H.; Elfalleh, W.; Laajel, M.; Ennajeh, I.; Mechlouch, R.F.; Nagaz, K. Chemical profiles and antioxidant activities of leaf, pulp, and stone of cultivated and wild olive trees (*Olea europaea* L.). *Int. J. Fruit. Sci.* **2020**, *20*, 350–370. [CrossRef]
34. Tufarelli, V.; Passantino, L.; Zupa, R.; Crupi, P.; Laudadio, V. Suitability of dried olive pulp in slow-growing broilers: Performance, meat quality, oxidation products, and intestinal mucosa features. *Poult. Sci.* **2022**, *101*, 102230. [CrossRef]
35. Dedousi, A.; Kotzamanidis, C.; Kritsa, M.Z.; Tsourekis, A.; Andreadelli, A.; Patsios, S.I.; Sossidou, E. Growth performance, gut health, welfare and qualitative behavior characteristics of broilers fed diets supplemented with dried common (*Olea europaea*) olive pulp. *Sustainability* **2023**, *15*, 501. [CrossRef]
36. Surai, P.F. Polyphenol compounds in the chicken/animal diet: From the past to the future. *J. Anim. Physiol. Anim. Nutr.* **2014**, *98*, 19–31. [CrossRef]
37. Muñoz-González, I.; Chamorro, S.; Pérez-Jiménez, J.; López-Andrés, P.; Álvarez-Acero, I.; Herrero, A.M.; Nardoia, M.; Brenes, A.; Viveros, A.; Arijia, I.; et al. Phenolic metabolites in plasma and thigh meat of chickens supplemented with grape byproducts. *J. Agric. Food Chem.* **2019**, *67*, 4463–4471. [CrossRef]
38. Abdel-Moneim, A.M.E.; Shehata, A.M.; Alzahrani, S.O.; Shafi, M.E.; Mesalam, N.M.; Taha, A.E.; Swelum, A.A.; Arif, M.; Fayyaz, M.; Abd El-Hack, M.E. The role of polyphenols in poultry nutrition. *J. Anim. Physiol. Anim. Nutr.* **2020**, *104*, 1851–1866. [CrossRef] [PubMed]
39. Agah, M.; Mirakzehi, M.T.; Saleh, H. Effects of olive leaf extract (*Olea europea* L.) on growth performance, blood metabolites and antioxidant activities in broiler chickens under heat stress. *J. Anim. Plant Sci.* **2019**, *29*, 657–666.
40. Elbaz, A.M.; Thabet, H.; Gad, G.; Mark, C. Productive and physiological performance of broilers fed diets containing different levels of olive pulp. *J. Anim. Poult. Prod.* **2020**, *11*, 435–439. [CrossRef]
41. Almuhayawi, M.S.; Alruhaili, M.H.; Gattan, H.S.; Alharbi, M.T.; Nagshabandi, M.K.; Almehayawi, M.S.; Jaouni, S.K.A.; Selim, S.; Alqahtani, F.S.; El-Saadony, M.T.; et al. Evaluation of antimicrobial effect of olive leaves powder and its role in improving the broiler productivity, carcass traits, blood metabolites, and caecal microbiota. *Poult. Sci.* **2023**, *102*, 103054. [CrossRef]
42. Albendea, P.; Guardiola, F.; Rafecas, M.; Vichi, S.; Barroeta, A.C.; Verdú, M.; Tres, A. Effect of feeding olive pomace acid oil on dark chicken meat lipid composition, oxidative stability, color, and sensory acceptance. *Animals* **2023**, *13*, 1343. [CrossRef]
43. Saleh, A.; Alzawqari, M. Effects of replacing yellow corn with olive cake meal on growth performance, plasma lipid profile, and muscle fatty acid content in broilers. *Animals* **2021**, *11*, 2240. [CrossRef] [PubMed]

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.